

Google Agent Development Kit - Build & Deploy AI Agents

Lab Setup Guide

Cloning the Course Repository and Setting Up Your Local Environment

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Overview

This guide walks you through cloning the course repository to your local machine. All demo code, lab starter files, and reference solutions are organized by module inside the repo. You will use this repository throughout every hands-on lab in the course.

Repository URL:

<https://github.com/janakiramm/google-adk-course>

Prerequisites

Before you begin, make sure the following tools are installed and available on your machine.

- [Git](#) 2.x or later
- [Python](#) 3.10 or later
- [pip](#) 23.x or later
- [Google Cloud SDK](#) (latest)
- [Google ADK](#) (latest)

Google Account required. You will need a Google account with access to a GCP project.

Step 1. Clone the Repository

Open your terminal and run the following command to clone the repository to your local machine.

```
git clone https://github.com/janakiramm/google-adk-course.git
```

Once the clone is complete, change into the project directory.

```
cd google-adk-course
```

Step 2. Verify the Contents

Run the command below to confirm all module folders are present.

```
ls -la
```

You should see a folder for each module.

Step 3. Set Up a Python Virtual Environment

It is strongly recommended to use a virtual environment to avoid dependency conflicts between modules.

Create and activate the environment

```
python3 -m venv .venv
```

```
# On macOS / Linux  
source .venv/bin/activate
```

```
# On Windows  
.venv\Scripts\activate
```

Install dependencies

Each module folder contains its own requirements.txt. Navigate into the module you are working on and install its dependencies.

```
cd Module-04  
pip install -r requirements.txt
```

Step 4. Authenticate with Google Cloud

The labs that use Vertex AI or Cloud Run require you to be authenticated with your Google Cloud account.

```
gcloud auth login  
gcloud config set project YOUR_PROJECT-ID  
gcloud config set compute/region us-central1
```

```
# Application default credentials for ADK and Vertex AI SDK  
gcloud auth application-default login
```

Application Default Credentials. The ADK and Vertex AI SDK both rely on Application Default Credentials (ADC). Running `gcloud auth application-default login` ensures that local ADK runs and SDK calls authenticate automatically.

Keeping Your Local Copy Up to Date

If the course repository is updated after you clone it, pull the latest changes with the following command from inside the repo folder.

```
git pull origin main
```

Troubleshooting

- `git: command not found` - Git is not installed. Install it from git-scm.com.
- `Permission denied (publickey)` - SSH key is not configured. Use the HTTPS clone URL shown in Step 1 instead.
- `ModuleNotFoundError` after installing - Wrong Python environment is active. Confirm your virtual environment is activated before running `pip install`.
- `adk: command not found` - ADK is not installed globally. Run `pip install google-adk` inside your activated virtual environment.
- `PERMISSION_DENIED` from Vertex AI - ADC is not configured or the wrong project is set. Re-run `gcloud auth application-default login` and verify your project ID.

Support and Resources

- Course Repository:
<https://github.com/janakiramm/google-adk-course>
- Google ADK Documentation:
<https://google.github.io/adk-docs/>
- Google ADK GitHub:
<https://github.com/google/adk-python>
- Vertex AI Documentation:
<https://cloud.google.com/vertex-ai/docs>